

ProGCo: Progressive Geometry-Coverage Co-Refinement for Sparse-View 3D Gaussian Splatting

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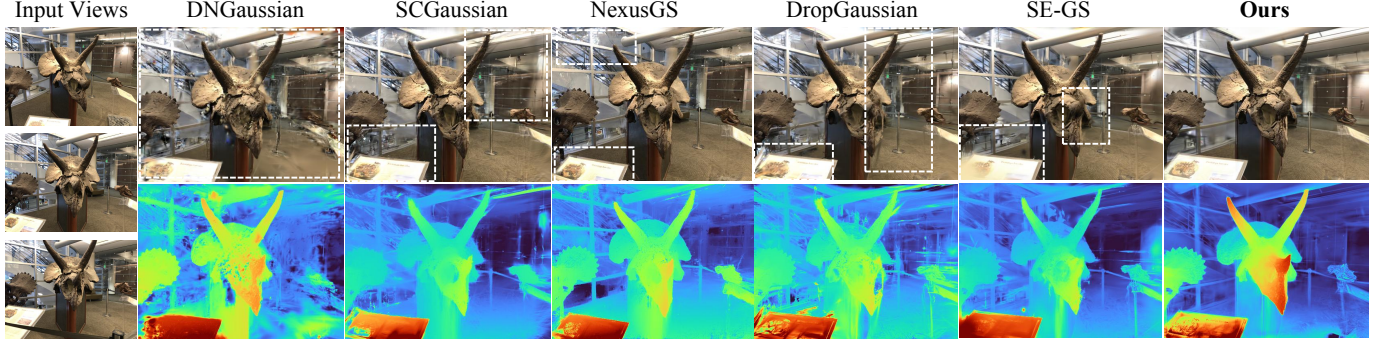


Fig. 1. Qualitative comparison of one synthesized novel view under the three-view setting. Compared with existing methods, our framework renders a higher-fidelity RGB image and more accurate depth map.

Abstract—Sparse novel view synthesis (NVS) with 3D Gaussian Splatting (3DGS) remains challenging, as limited input views yield unreliable geometry that induces severe floaters and background collapse in rendered images. Existing methods rely on patch-level depth regularization or generative scene expansion; however, patch-based depth constraints disrupt structural integrity of individual objects, while unconstrained generative images lack 3D geometric consistency. To address these limitations, we propose ProGCo, a progressive geometry-coverage co-refinement framework that establishes closed-loop mutual reinforcement between geometric refinement and scene expansion. Rather than treating Gaussian positioning and scene expansion independently, ProGCo first restores multi-view coherent geometry by preserving intra-object structural integrity and global spatial layout. Guided by this refined scaffold, ProGCo progressively interpolates geometry-consistent pseudo-views to expand scene coverage, which in turn drives finer geometric optimization. Extensive experiments across forward-facing and unbounded 360° benchmarks demonstrate that ProGCo consistently performs favorably against existing state-of-the-art methods.

Index Terms—Sparse novel view synthesis, 3D Gaussian splatting, progressive scene expansion.

I. INTRODUCTION

SPARSE novel view synthesis (NVS) aims to reconstruct photorealistic scenes and render novel views from sparse input images. As a foundational task in computer vision and graphics, NVS empowers a wide range of downstream applications, including autonomous driving, augmented reality, and robotics. Recently, 3DGS [1] has emerged as a leading paradigm for real-time, high-fidelity rendering. Standard

3DGS relies heavily on Structure-from-Motion (SfM) methods like COLMAP [2] to initialize Gaussian centers from sparse point clouds. In dense-view settings, wide view overlap and high multi-view redundancy facilitate robust feature matching and precise triangulation, yielding a reliable geometric basis. Under sparse views, however, these spatial cues deteriorate significantly, resulting in severe degradation of point cloud density and accuracy. Consequently, optimizing 3DGS under sparse-view conditions remains a critical challenge.

Starting from an unreliable point cloud initialization, sparse RGB supervision provides only weak constraints on Gaussian positions. Consequently, inaccurately positioned primitives may fit certain views well while yielding less accurate representations in others. Moreover, the adaptive density control scheme of 3DGS locally clones or splits existing Gaussian primitives based on accumulated pixel-space gradients. This scheme exacerbates initial positional errors while failing to recover regions poorly covered by the sparse point cloud. These two issues form a tightly coupled dilemma: imprecise Gaussian positions lead to unreliable scene expansion, whereas insufficient spatial coverage fails to effectively correct primitive positions. We observe that this coupled problem hinders reliable geometric reconstruction, causing severe floaters and background collapse [3] in rendered images. To address this challenge, we jointly correct Gaussian positions and expand scene coverage to reconstruct scene geometry reliably.

Existing sparse-view 3DGS methods typically address these two coupled issues separately. One line of approaches [3]–[6] improves Gaussian positional accuracy using monocular depth priors. By regularizing rendered depth maps via monocular predictions within local patches, these approaches alleviate localized primitive misplacement. However, such patch-level supervision often fragments individual objects across multiple patches, neglecting object-level geometric integrity. Consequently, Gaussians belonging to the same object may be

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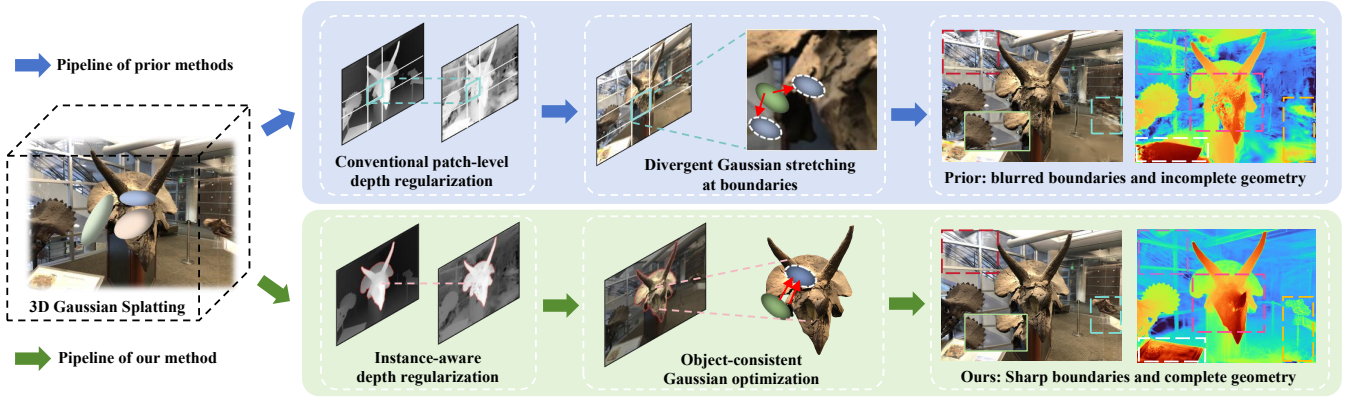


Fig. 2. **Comparison of Gaussian deformation near boundaries.** Conventional patch-level methods (e.g., DNGaussian [4], SparseGS [3]) often inconsistently stretch Gaussian ellipsoids at object boundaries and patch interfaces, resulting in blurred edges and incomplete structures. In contrast, our instance-aware depth regularization consistently guides primitive movement within individual objects, preserving geometric integrity of objects and sharp boundaries.

pulled in conflicting directions at patch boundaries, leading to incomplete structures and blurry boundaries, as illustrated in Fig. 2. While object-level depth supervision offers a natural remedy, off-the-shelf segmentors like SAM [7] may struggle to delineate precise masks in complicated scenes. In addition, recent methods [8], [9] increase initial point cloud density using epipolar constraints and co-visibility maps. While effective in observed regions, these methods remain limited in recovering unobserved scene structures. A second line of work [10]–[14] leverages generative video models to expand spatial coverage of Gaussian primitives. Despite producing visually compelling sequences, these methods lack explicit 3D geometric constraints during generation, potentially causing structural artifacts that compromise geometric consistency.

To address the coupled challenge, we present ProGCo, a Progressive Geometry-Coverage Co-refinement framework that jointly corrects Gaussian positions and expands scene coverage for high-fidelity NVS. Our key insight is that positional correction and spatial expansion cannot be naively decoupled or solved in a single step. Faithful scene expansion requires a reliable geometric basis, as pseudo-views rendered from inaccurate primitives can amplify structural errors. Moreover, directly expanding scene coverage to distant unseen viewpoints often leaves large regions weakly constrained, producing severe Gaussian holes that further degrade the geometry. To break this error propagation loop, ProGCo adopts a progressive two-stage schedule:

1. Instance-Aware Geometric Coherence Refinement.

ProGCo first corrects Gaussian positions before expanding coverage. Rather than regularizing arbitrary image patches, we introduce an instance-aware depth constraint to guide Gaussian updates within each instance segment. By treating each segment as a coherent object, we jointly refine the positions of its associated 3D Gaussians across views, thereby promoting consistent geometry for the same physical structures. This constraint preserves object-level geometric integrity and sharp boundaries, as illustrated in Fig. 2. However, in large-scale scenes with complex textures, occlusions may cause inaccurate segmentation, leaving some Gaussians unconstrained. To handle these segmentation failures and preserve the global

spatial layout, we further enforce scale-invariant alignment over the full depth map to refine the positions of all primitives. Together, these complementary constraints establish a robust geometric scaffold for subsequent scene expansion.

2. Progressive Geometry-Grounded Scene Expansion.

Rather than directly expanding coverage to distant viewpoints, where large unobserved regions may produce severe floaters and Gaussian holes that corrupt the geometry, ProGCo expands scene coverage iteratively. We progressively interpolate new camera poses from proximal to distal viewpoints, with each interpolation step building on the geometry refined at the preceding stage. To support diverse camera trajectories, our adaptive interpolation scheme accommodates both forward-facing and 360° unbounded scenes. To prevent residual artifacts in interpolated pseudo-views from compromising geometric consistency, we employ a pretrained diffusion model to suppress floaters and retain only pseudo-views with high structural similarity to the raw renderings. The accepted pseudo-images further foster subsequent geometry refinement.

This staged co-refinement process forms a virtuous cycle: refined geometry produces higher-fidelity pseudo-views, which in turn facilitates subsequent geometry refinement. Extensive evaluations on four real-world benchmarks, including LLFF [15], Tanks and Temples [16], IBRNet [17], and Mip-NeRF360 [18], demonstrate the superior performance of ProGCo (Fig. 1). Our main contributions are summarized as:

- We propose **ProGCo**, a progressive geometry-coverage co-refinement framework for sparse-view NVS that addresses the coupled bottleneck of imprecise Gaussian positions and limited spatial coverage.
- We introduce an instance-aware geometry refinement that establishes coherent geometry across views by preserving intra-object structural integrity and global scene layout.
- We design progressive scene expansion from proximal to distal viewpoints, using diffusion priors to suppress floaters while retaining faithful pseudo-view supervision.

II. RELATED WORK

Sparse Novel View Synthesis. Sparse NVS aims to synthesize unobserved views from extremely sparse inputs. This

is an inherently ill-posed 3D reconstruction problem. While Neural Radiance Fields (NeRF) [19] and 3DGS [1] deliver photorealistic rendering under dense input coverage, their performance degrades severely in sparse-view settings due to the under-constrained nature of the problem. To address this challenge, existing NeRF-based methods incorporate auxiliary constraints, such as monocular depth distillation [20]–[22], geometric regularization [23], [24], and explicit grid or voxel representations for improved efficiency [25]–[30]. However, despite recovering reasonable geometry, NeRF-based frameworks remain bottlenecked by prolonged training and rendering latency.

To enable real-time, high-fidelity rendering, recent work increasingly leverages explicit 3DGS representations [31]–[41]. These approaches rely heavily on geometric priors to constrain Gaussian optimization, drawing on cross-view motion cues [9], [42], [43], monocular depth predictions [3], [4], [44], [45], or pose-free alignment formulations [37], [38], [46]–[48]. Nevertheless, high-fidelity reconstruction requires both precise primitive positioning and comprehensive scene coverage. Current sparse 3DGS methods focus primarily on regularizing Gaussian positions within observed regions or enhancing rendering robustness. However, unobserved regions remain severely under-represented, leaving unaddressed the core dilemma of establishing reliable geometry when positional accuracy and spatial coverage are jointly deficient.

Monocular depth priors from estimators [49]–[51] are widely deployed to regularize sparse-view neural rendering. While early NeRF-based approaches [21], [23] typically reconstruct absolute depth, contemporary monocular depth models predict relative depth, which is subject to inherent scale and shift ambiguities. To mitigate these ambiguities, existing sparse-view 3DGS methods incorporate scale-invariant losses [3], [23] or cross-view consistency constraints [52]. However, globally optimizing a unified scale factor often oversmooth fine-grained local structures. While methods [4] and [3] address this issue via joint local and global depth regularization, their reliance on fixed-size patches neglects the geometric integrity of objects, causing Gaussian misplacement and thus inducing geometric tearing, depth disorder, and boundary blur. Although such depth regularization improves local positional accuracy, it cannot recover unobserved regions absent from the input views. To address these challenges, we incorporate scale-invariant instance-aware depth regularization and scene-level depth alignment to construct a coherent geometric basis for scene expansion.

Generative Priors for Sparse NVS. Generative models are increasingly utilized to expand scene coverage by synthesizing additional images. Recent diffusion-based methods [11], [12], [53]–[55] often fine-tune video diffusion models to generate image sequences. Despite high visual quality, these images suffer from structural inconsistencies due to the absence of 3D geometric consistency constraints during generation. Other methods [56]–[58] are dedicated to reconstructing single objects rather than complex, unbounded real-world environments.

To bridge this gap, recent approaches [13], [59]–[61] integrate generative priors to rectify under-constrained 3DGS

representations. Notably, DIFIX3D+ [59] fine-tunes a single-step diffusion model [62] to improve the quality of pseudo-views; however, its expansion strategy assumes a highly dense and structurally sound initial geometry, making it unsuitable for extreme sparse-view settings. In contrast, our proposed method, **ProGCo**, leverages the pretrained diffusion model of DIFIX3D+ strictly as a generative filter to suppress floaters in rendered views. Crucially, candidate pseudo-views are integrated only when they exhibit high structural similarity to the original renderings, preventing generative hallucination from corrupting scene optimization.

III. METHODOLOGY

A. Overview

As illustrated in Fig. 3, ProGCo consists of a geometry-refinement stage and a progressive pseudo-view interpolation stage. Starting from a sparse point cloud initialized by [2], ProGCo first corrects the Gaussian positions to establish a coherent geometric scaffold across views. Using the geometry refined in the preceding stage, ProGCo then progressively interpolates pseudo-views to expand scene coverage faithfully. The resulting pseudo-images provide additional supervision for geometry optimization in the next stage. This staged process forms a co-refinement cycle in which geometric consistency and scene coverage progressively reinforce each other. Algorithmic pseudocode is provided in Section I of the supplementary material.

Formally, given a sparse input set $\mathcal{D}^{\text{spa}} = \{(I_i, D_i^{\text{mono}}, P_i)\}_{i=1}^N$, where I_i , D_i^{mono} , and $P_i = (\mathbf{R}_i, \mathbf{t}_i)$ denote the RGB image, monocular depth map, and camera pose of the i -th input, respectively, our framework optimizes a 3DGS representation $\mathcal{G}_\Theta$ to synthesize high-fidelity novel views of real-world scenes.

B. Instance-Aware Geometric Coherence Refinement

To correct misplaced 3D Gaussians, existing approaches typically utilize monocular depth priors and multi-view matching cues, such as co-visibility maps and epipolar constraints. However, depth-regularized methods often supervise local patches independently, potentially disrupting object-level structural integrity and causing geometric inconsistencies across views. Moreover, matching priors are typically constructed between pairs of input views and do not explicitly enforce global structural coherence across multiple views. To overcome these limitations, we propose a geometric coherence refinement strategy that performs scale-invariant depth alignment at both the instance and scene levels, thereby improving cross-view geometric consistency.

Instance-Aware Depth Regularization. To address this issue, we propose instance-aware depth regularization that constrains object-level structural integrity using monocular depth priors. Given an RGB image $I_i \in \mathbb{R}^{3 \times H \times W}$ and its monocular depth map $D_i^{\text{mono}} \in \mathbb{R}^{H \times W}$ estimated by Depth Anything V2 (DA2) [51], 3DGS projects the 3D Gaussian primitives onto the image plane. For a pixel (x, y) , let $\mathcal{N}_i(x, y)$ denote the overlapping Gaussians ordered from front to back. The radial depth of the k -th Gaussian is defined as $d_{i,k} = \|\boldsymbol{\mu}_k - \mathbf{o}_i\|_2$,

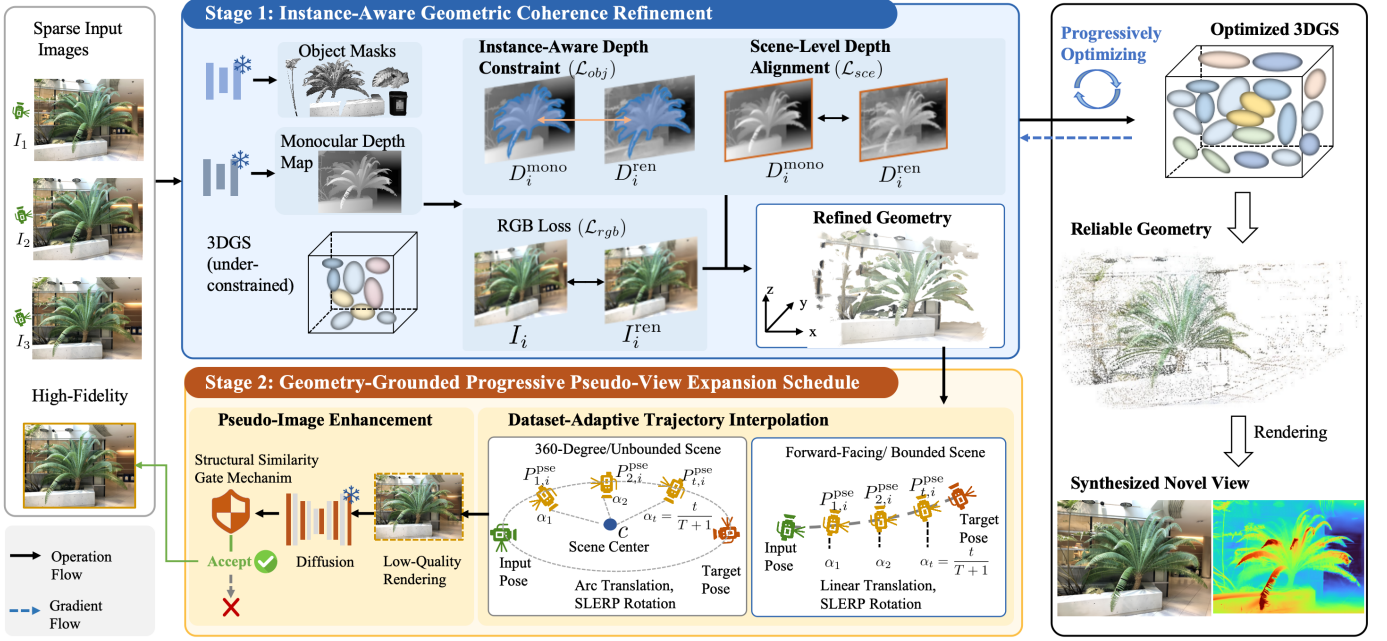


Fig. 3. **Overview of the ProGCo pipeline.** In the first stage, instance-aware geometric coherence refinement establishes a geometry scaffold that preserves object-level structural integrity across views and maintains consistent inter-object spatial relationships. In the second stage, we progressively interpolate pseudo-images to reliably expand scene coverage based on the refined geometry. This staged co-refinement process forms a virtuous cycle that enables high-fidelity NVS.

where μ_k and $\mathbf{o}_i$ denote the Gaussian center and camera center, respectively. The accumulated transmittance before the k -th Gaussian is defined as

$$T_{i,k}(x, y) = \prod_{\substack{j \in \mathcal{N}_i(x, y) \\ j < k}} (1 - \tilde{\alpha}_{i,j}(x, y)). \quad (1)$$

Following the alpha-composited depth formulation in [4], the rendered radial depth at (x, y) is

$$D_i^{\text{ren}}(x, y) = \sum_{k \in \mathcal{N}_i(x, y)} T_{i,k}(x, y) \tilde{\alpha}_{i,k}(x, y) d_{i,k}. \quad (2)$$

Here, $\tilde{\alpha}_{i,k}(x, y)$ denotes the effective opacity of the projected k -th Gaussian at pixel (x, y) . Collecting these pixel-wise depths yields the rendered depth map $D_i^{\text{ren}} \in \mathbb{R}^{H \times W}$. Before computing the depth losses, both D_i^{ren} and D_i^{mono} are normalized to the range $[0, 1]$. The rendered RGB image I_i^{ren} is obtained using the standard differentiable 3DGS renderer.

To refine Gaussian positions at the object level, we impose scale-invariant depth constraints within instance regions in the rendered depth map. Specifically, we employ the Segment Anything Model (SAM) [7] to extract a set of instance masks $\mathcal{M} = \{M_l\}_{l=1}^L$ from the input image I_i . Each mask $M_l \in \{0, 1\}^{H \times W}$ is a binary map with 1 indicating the pixels belonging to the l -th object, and 0 indicating the background. Although SAM independently predicts the masks for each input view, we empirically observe that it usually delineates the same physical objects as coherent regions across views. We do not need to compute explicit cross-view instance correspondences. Instead, instance-wise depth constraints from all training views jointly optimize the shared 3D Gaussian representation, implicitly encouraging Gaussians associated with the same physical object to receive compatible geometric

updates across views. The instance-aware depth regularization loss is defined as

$$\mathcal{L}_{\text{obj}} = \frac{1}{L} \sum_{l=1}^L [1 - \rho(D_i^{\text{mono}} \odot M_l, D_i^{\text{ren}} \odot M_l)], \quad (3)$$

where $\odot$ denotes the Hadamard product, and $\rho(X, Y)$ is the Pearson Correlation Coefficient (PCC):

$$\rho(X, Y) = \frac{\sum_p (X_p - \bar{X})(Y_p - \bar{Y})}{\sqrt{\sum_p (X_p - \bar{X})^2} \sqrt{\sum_p (Y_p - \bar{Y})^2 + \epsilon}}. \quad (4)$$

Consequently, Gaussian primitives belonging to the same object are consistently optimized under object-coherent depth constraints. Moreover, the scale invariance of PCC removes the need for metric alignment between monocular and rendered depths, further improving geometric consistency across views.

Scene-Level Depth Alignment. Although instance-level supervision preserves object structure, SAM may produce inaccurate masks in occluded regions or miss small background structures, leaving the corresponding Gaussians insufficiently constrained. More importantly, the instance-level depth loss aligns each instance independently and thus does not explicitly preserve inter-object spatial relationships. Because PCC is invariant to positive affine transformations, i.e., $\rho(X, \alpha Y + \beta) = \rho(X, Y)$, $\alpha > 0$, independently optimized objects may undergo different scale and shift transformations, resulting in inconsistent inter-object depth relationships. To address segmentation failures and depth ambiguity between objects, we introduce complementary scene-level depth regularization over the full depth map:

$$\mathcal{L}_{\text{scene}} = 1 - \rho(D_i^{\text{mono}}, D_i^{\text{ren}}). \quad (5)$$

TABLE I
Quantitative comparisons on LLFF, IBRNet, and T&T with three input views. The best, second-best, and third-best results are highlighted in red, orange, and yellow. * indicates results reproduced using the official code.

Category	Method	LLFF (3 views)			IBRNet (3 views)			Tanks (3 views)		
		PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
NeRF-based	Mip-NeRF [63]	14.62	0.351	0.495	15.83	0.406	0.488	12.57	0.241	0.623
	RegNeRF [64]	19.08	0.587	0.336	19.05	0.542	0.377	13.12	0.268	0.618
	FreeNeRF [65]	19.63	0.612	0.308	19.76	0.588	0.333	12.30	0.308	0.636
	SparseNeRF [23]	19.86	0.624	0.328	19.90	0.593	0.364	13.66	0.331	0.615
3DGS-based	3DGS [1]	18.54	0.588	0.272	17.79	0.538	0.377	17.14	0.493	0.397
	FSGS [6]	20.43	0.682	0.248	19.84	0.648	0.306	20.01	0.652	0.323
	DNGaussian* [4]	19.12	0.591	0.294	19.03	0.620	0.371	18.02	0.582	0.413
	AD-GS [66]	20.06	0.699	0.237	-	-	-	19.72	0.678	0.285
	SE-GS* [34]	20.60	0.721	0.187	20.27	0.716	0.549	18.96	0.693	0.294
	DropGaussian* [67]	20.32	0.705	0.204	20.42	0.697	0.548	18.27	0.655	0.329
	SCGaussian* [68]	20.41	0.705	0.218	21.59	0.731	0.233	20.98	0.703	0.303
	NexusGS* [8]	21.07	0.738	0.177	-	-	-	-	-	-
	ProGCo (Ours)	22.06	0.740	0.177	22.41	0.752	0.298	21.48	0.707	0.285

This global constraint aligns the rendered depth with the monocular depth over the entire image, preserving inter-object spatial relationships while supervising regions not covered by instance masks. Together, the instance- and scene-level depth constraints establish a cross-view coherent geometry.

C. Progressive Geometry-Grounded Scene Expansion

After establishing a coherent geometry scaffold, we expand scene coverage to regions poorly represented by the sparse initial point cloud. Direct extrapolation to distant unseen viewpoints, however, may produce unfaithful scene content, amplifying residual errors and compromising the established geometric consistency. We therefore adopt a progressive geometry-grounded scene expansion strategy that interpolates pseudo-views from proximal to distal viewpoints, enabling the rendering of pseudo-views that builds upon the consistent geometry refined in the preceding stage.

Specifically, the optimization schedule is divided into T stages defined by a sequence of interpolation steps $\mathcal{S} = \{s_t\}_{t=1}^T$. A fixed set of target camera poses is first selected to encourage broader scene coverage. At step s_t , each pseudo-view is interpolated between an input viewpoint and its corresponding target, progressively extending the synthesized views from proximal to distal regions. We render the current 3DGS representation from these interpolated viewpoints to produce pseudo-images, which are refined and filtered before being incorporated into the training set. The accepted images provide additional supervision for geometry optimization, enabling the subsequent stage to expand from a more reliable geometry. This iterative process establishes a geometry-coverage co-refinement loop, in which progressively refined geometry supports increasingly broader scene expansion.

Target View Selection. Given the sparse training poses $\mathcal{P}^{\text{spa}}$ and a candidate set of held-out poses $\mathcal{P}^{\text{cand}}$, we select target poses that are spatially distant from the training poses. For each candidate pose $\mathbf{P}_j^{\text{cand}}$, we define its novelty score as the

Euclidean distance between its camera center $\mathbf{o}_j^{\text{cand}}$ and the nearest input camera center:

$$d_j = \min_{\mathbf{P}_i^{\text{spa}} \in \mathcal{P}^{\text{spa}}} \|\mathbf{o}_j^{\text{cand}} - \mathbf{o}_i^{\text{spa}}\|_2. \quad (6)$$

A larger d_j indicates a greater distance from the training poses. We rank all candidate poses by d_j and retain the K ($K \leq N$) highest-scoring poses as fixed targets:

$$\mathcal{P}^{\text{tar}} = \{\mathbf{P}_j^{\text{cand}} \mid d_j \in \text{TopK}(\{d_m\}_{m=1}^M, K)\}. \quad (7)$$

Trajectory-Adaptive Pseudo-View Interpolation. At interpolation step s_t , we interpolate N_{pse} pseudo-poses between sparse input poses and their corresponding target poses in $\mathcal{P}^{\text{tar}}$, using an interpolation fraction $\alpha_t = \frac{t}{T+1}$, where $t \in \{1, \dots, T\}$. Since $\alpha_t < 1$ at every stage, all pseudo-poses remain strictly between the input and target poses. The target poses serve only as interpolation anchors and are never sampled during training.

To generate valid pseudo-views for diverse camera trajectories, we propose a trajectory-adaptive pose interpolation strategy. For forward-facing or bounded scenes, we linearly interpolate camera translations and apply spherical linear interpolation (SLERP) [69] to camera rotations:

$$\mathbf{t}_{t,i}^{\text{pse}} = (1 - \alpha_t)\mathbf{t}_i^{\text{spa}} + \alpha_t\mathbf{t}_i^{\text{tar}}, \quad \mathbf{R}_{t,i}^{\text{pse}} = \text{SLERP}(\mathbf{R}_i^{\text{spa}}, \mathbf{R}_i^{\text{tar}}, \alpha_t). \quad (8)$$

For unbounded 360° scenes, linear translation interpolation may cause the camera trajectory to pass through the scene interior, resulting in invalid viewpoints. We therefore interpolate each camera pose along an arc around the scene center $\mathbf{c}$ estimated from the sparse input views. Let $\hat{\mathbf{t}}^q = (\mathbf{t}^q - \mathbf{c}) / \|\mathbf{t}^q - \mathbf{c}\|_2$ and $r^q = \|\mathbf{t}^q - \mathbf{c}\|_2$, where $q \in \{\text{spa}, \text{tar}\}$, denote the normalized direction and radial distance of a camera pose relative to $\mathbf{c}$, respectively. The translation is given by

$$\mathbf{t}_{t,i}^{\text{pse}} = \mathbf{c} + [(1 - \alpha_t)r_i^{\text{spa}} + \alpha_t r_i^{\text{tar}}] \text{SLERP}(\hat{\mathbf{t}}_i^{\text{spa}}, \hat{\mathbf{t}}_i^{\text{tar}}, \alpha_t). \quad (9)$$

TABLE II
Quantitative comparisons on Mip-NeRF360 with 12 and 24 input views.

Method	12-view			24-view		
	PSNR(↑)	SSIM(↑)	LPIPS (↓)	PSNR(↑)	SSIM(↑)	LPIPS (↓)
3DGS [1]	18.52	0.523	0.415	22.80	0.708	0.276
FSGS [6]	18.80	0.531	0.418	23.70	0.745	0.230
CoR-GS [32]	19.52	0.558	0.418	23.39	0.727	0.271
NexusGS [8]	-	-	-	23.86	0.753	0.206
AD-GS [66]	19.66	0.593	0.386	23.68	0.750	0.249
SE-GS* [34]	19.91	0.596	0.400	23.74	0.745	0.265
DropGaussian* [67]	20.17	0.605	0.337	24.13	0.762	0.225
ProGCo (Ours)	21.40	0.645	0.294	24.88	0.779	0.203

The camera rotation follows the same SLERP rule as in Eq. (8). This arc-based interpolation smoothly varies both the angular direction and radial distance, reducing the risk of generating invalid camera poses inside the scene. The resulting pseudo-pose set is $\mathcal{P}_t^{\text{pse}} = \{\mathbf{P}_{t,i}^{\text{pse}}\}_{i=1}^{N_{\text{pse}}}$, where $\mathbf{P}_{t,i}^{\text{pse}} = (\mathbf{R}_{t,i}^{\text{pse}}, \mathbf{t}_{t,i}^{\text{pse}})$. The 3DGS model then renders the corresponding pseudo-images and depth maps from these interpolated poses:

$$I_{t,i}^{\text{pse}}, D_{t,i}^{\text{pse,ren}} = \mathcal{G}_{\Theta}(\mathbf{P}_{t,i}^{\text{pse}}), \quad \text{for } i = 1, 2, \dots, N_{\text{pse}}. \quad (10)$$

Reliable Pseudo-View Refinement. The rendered pseudo-images $\{I_{t,i}^{\text{pse}}\}_{i=1}^{N_{\text{pse}}}$ may contain visible artifacts in expanded regions when the underlying scene geometry remains insufficiently refined. To alleviate these artifacts, we employ DiFix [59], a diffusion-based refiner pretrained on large-scale datasets. Given a raw pseudo-image $I_{t,i}^{\text{pse}}$ and its nearest reference image $I_{t,i}^{\text{ref}}$, the refinement module Φ_{diff} yields

$$\hat{I}_{t,i}^{\text{pse}} = \Phi_{\text{diff}}(I_{t,i}^{\text{pse}}, I_{t,i}^{\text{ref}}), \quad (11)$$

where $I_{t,i}^{\text{ref}}$ serves as guidance to synthesize plausible and contextually consistent content within artifact-corrupted regions. Although this refinement enhances visual quality, it lacks explicit 3D geometric constraints and may introduce structural inconsistencies. We therefore filter the refined pseudo-images based on their structural similarity to the raw renderings:

$$q_{t,i} = \text{SSIM}(I_{t,i}^{\text{pse}}, \hat{I}_{t,i}^{\text{pse}}), \quad a_{t,i} = \mathbb{I}[q_{t,i} \geq \eta]. \quad (12)$$

Only pseudo-images satisfying $a_{t,i} = 1$ are retained, providing faithful scene expansion while preventing noisy structure from corrupting the underlying geometry. For each accepted pseudo-image, its depth map is estimated using DA2 [51]:

$$\hat{D}_{t,i}^{\text{pse}} = \Psi_{\text{depth}}(\hat{I}_{t,i}^{\text{pse}}), \quad (13)$$

and the results are accumulated as follows:

$$\bar{\mathcal{D}}^{\text{pse}} \leftarrow \bar{\mathcal{D}}^{\text{pse}} \cup \{(\hat{I}_{t,i}^{\text{pse}}, \hat{D}_{t,i}^{\text{pse}}, P_{t,i}^{\text{pse}}) \mid a_{t,i} = 1\}. \quad (14)$$

The training set is subsequently augmented using $\mathcal{D}^{\text{train}} = \mathcal{D}^{\text{spa}} \cup \bar{\mathcal{D}}^{\text{pse}}$, which provides additional supervision for previously unobserved regions and promotes subsequent geometric optimization. In turn, the improved geometry enables 3DGS to generate higher-quality pseudo-images from increasingly distant, unseen viewpoints. This iterative co-refinement between geometry and spatial coverage progressively reinforces the 3D representation, ultimately achieving high-fidelity NVS.

TABLE III
Quantitative comparisons on LLFF with 6 and 9 input views.

Method	6-view			9-view		
	PSNR(↑)	SSIM(↑)	LPIPS (↓)	PSNR(↑)	SSIM(↑)	LPIPS (↓)
Mip-NeRF [63]	22.91	0.756	0.213	24.88	0.826	0.170
RegNeRF [64]	23.10	0.760	0.206	24.86	0.820	0.161
FreeNeRF [65]	23.73	0.779	0.195	25.13	0.827	0.160
3DGS [1]	22.57	0.757	0.182	24.02	0.799	0.151
FSGS [6]	23.25	0.776	0.185	24.56	0.815	0.151
DNGaussian* [4]	22.18	0.755	0.198	23.17	0.788	0.180
CoR-GS [32]	23.33	0.780	0.187	24.75	0.817	0.152
AD-GS [66]	23.54	0.793	0.170	25.02	0.830	0.142
DropGaussian* [67]	24.41	0.829	0.128	25.13	0.849	0.119
SE-GS* [34]	24.65	0.838	0.111	26.36	0.878	0.084
ProGCo (Ours)	24.45	0.814	0.098	25.51	0.849	0.088

Training Objective. The overall joint optimization objective is defined as:

$$\mathcal{L} = \mathcal{L}_{\text{rgb}} + \lambda_o \mathcal{L}_{\text{obj}} + \lambda_s \mathcal{L}_{\text{scene}}, \quad (15)$$

where λ_o and λ_s control the contributions of the object- and scene-level regularization terms, respectively. The image reconstruction loss $\mathcal{L}_{\text{rgb}}$ follows the standard 3DGS formulation:

$$\mathcal{L}_{\text{rgb}} = (1 - \lambda_{\text{ssim}}) \|I_i^{\text{ren}} - I_i\|_1 + \lambda_{\text{ssim}} (1 - \text{SSIM}(I_i^{\text{ren}}, I_i)). \quad (16)$$

IV. EXPERIMENT

A. Experimental Settings

Datasets and Metrics. We evaluate ProGCo on four real-world benchmarks encompassing intricate textures and challenging illumination conditions. Specifically, LLFF [15], IBR-Net [17], and Tanks & Temples (T&T) [16] comprise scenes captured with forward-facing camera trajectories, whereas Mip-NeRF360 [18] contains unbounded 360° environments. Following established evaluation protocols [4], [34], [67], we report three standard metrics: Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), and Learned Perceptual Image Patch Similarity (LPIPS).

Implementation Details. We use pretrained DA2 [51] and SAM (ViT-B/16) [7] to estimate monocular depth maps and extract instance masks, respectively. We optimize the 3DGS model for 14,000 iterations on Mip-NeRF360 and 12,000 iterations on each of the other benchmarks. To stabilize geometric optimization, we perform pseudo-view interpolation at iterations $\{1500, 3500, 5500, 7500\}$. To balance the loss terms, we set $\lambda_o = 0.1, \lambda_s = 0.2$ in Eq. (15). ProGCo requires approximately 6.5 hours of training on a single A800 GPU while maintaining real-time inference with a runtime of 0.004s. Dataset configurations and baseline methods are provided in Section IV of the supplementary material.

B. Performance Evaluation

Quantitative Results. We comprehensively evaluate ProGCo for novel view synthesis and reconstruction under diverse sparse-input settings. *Results on Forward-Facing Scenes with Three Input Views.* We first evaluate ProGCo under the challenging three-view setting, in which sparse SfM initialization results in inaccurate Gaussian positions and insufficient scene

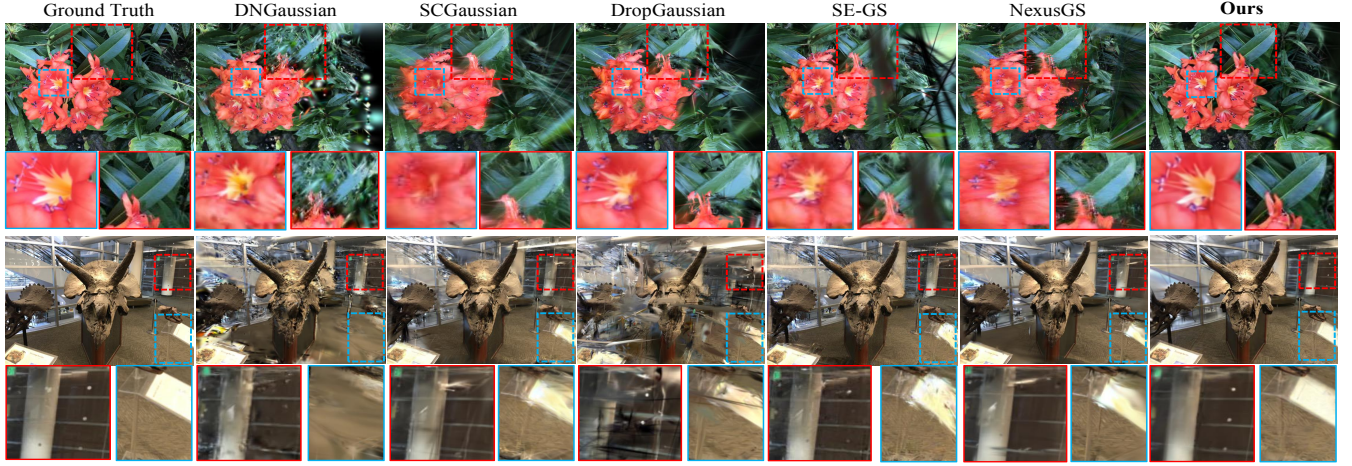


Fig. 4. **Qualitative comparisons on the LLFF dataset (3 views).** ProGCo reconstructs finer foreground and background details, including flower stamens and other thin structures.

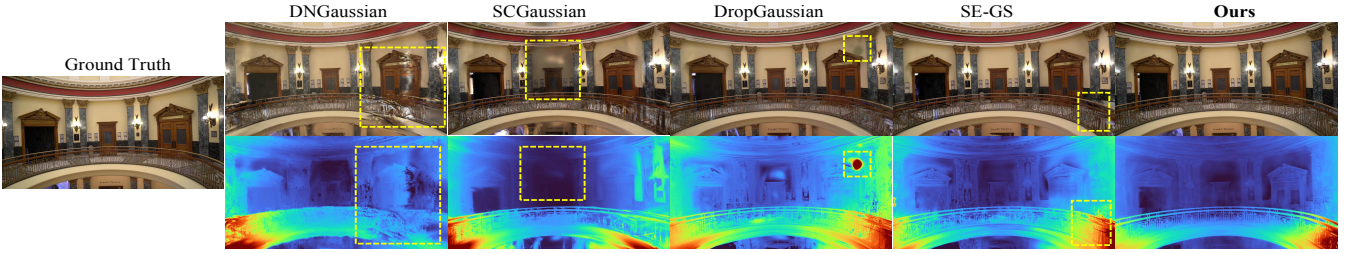


Fig. 5. **Qualitative comparisons on the T&T dataset (3 views).** The figure shows RGB renderings and depth maps. ProGCo renders higher-fidelity images with complex textures while preserving more accurate spatial layouts and structural details.

coverage. As shown in Table I, ProGCo achieves superior performance across all three forward-facing benchmarks. Its clear improvement over DNGaussian [4] highlights the advantage of instance-aware geometric coherence refinement over patch-wise depth regularization. ProGCo also outperforms SCGaussian [68] and NexusGS [8], demonstrating that refining geometry only within observed views is insufficient for high-fidelity NVS. Its superior performance relative to FSGS [6] and SE-GS [34] further suggests that these gains stem from the joint refinement of geometry and coverage rather than initial point-cloud densification or regularization alone. Overall, these results confirm that jointly improving Gaussian positions and scene coverage substantially enhances reconstruction quality under extremely sparse input conditions.

Results on Unbounded Scenes with 12 and 24 Input Views. We further evaluate our method on the unbounded scenes of the Mip-NeRF360 dataset. The 360° camera trajectories yield limited overlap among the input views and leave large regions unobserved, making reliable view expansion particularly challenging. As shown in Table II, ProGCo achieves state-of-the-art performance across all metrics under both input-view settings. DropGaussian [67] and SE-GS [34] primarily alleviate sparse-view overfitting through structural regularization and self-ensembling, respectively. In contrast, ProGCo synthesizes additional views to expand spatial coverage. These results demonstrate that our adaptive pseudo-view interpolation reliably expands unobserved regions in 360° scenes.

Scaling with Additional Input Views. We further evaluate

ProGCo on LLFF using 6 and 9 input views, as reported in Table III. Given the restricted viewing range of forward-facing scenes, using six or nine input views instead of three yields substantially greater inter-view overlap, thereby providing denser image-based supervision for scene reconstruction. Under these settings, SE-GS [34] achieves the best PSNR and SSIM, benefiting from self-ensembling regularization that mitigates overfitting via dense real-view supervision. DropGaussian [67] remains a strong baseline through stochastic Gaussian pruning during training. Nevertheless, ProGCo consistently ranks among the top methods, achieving the best LPIPS in the 6-view setting and strong results across all metrics. Compared with these regularization-based methods, ProGCo better preserves fine-grained structures by preserving geometric coherence across views, demonstrating its robustness as input-view coverage increases.

Qualitative Evaluation. We present extensive visual comparisons of rendered RGB images and depth maps. Additional results, including interpolated pseudo-views, computational efficiency analyses, and a video demo, are provided in Section II of the supplementary material.

Visual Comparisons on Novel View Synthesis. As illustrated in Fig. 1, ProGCo synthesizes a novel view with noticeably higher fidelity and recovers more coherent geometry than competing methods under the three-view setting, despite complex textures and illumination. Our framework preserves fine RGB details, such as complete museum labels and sharp object boundaries. Moreover, the rendered novel depth map

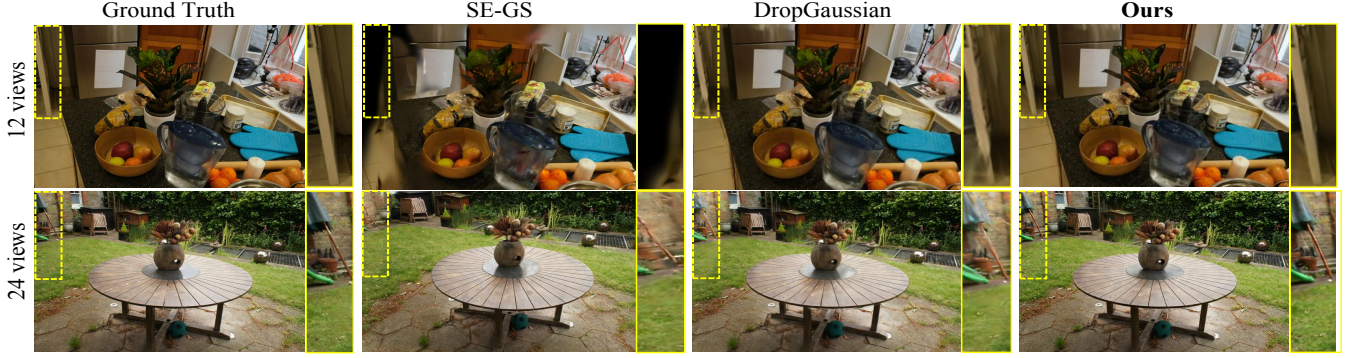


Fig. 6. **Qualitative comparisons on unbounded Mip-NeRF360 with 12 and 24 input views.** ProGCo better preserves thin background structures and produces visually coherent details.

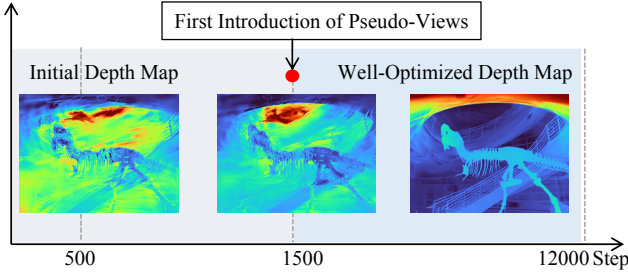


Fig. 7. **Depth evolution under the progressive pseudo-view schedule.** After initial geometric refinement, pseudo-views are introduced and further improve geometry, yielding a more accurate depth map.

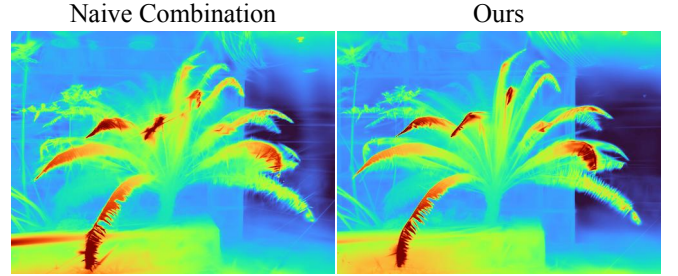


Fig. 8. **Naive combination versus progressive co-refinement.** The progressive schedule preserves finer structures and reduces local geometric distortions.

accurately captures inter-object spatial relationships while maintaining the complete object-level structures. These results confirm that jointly refining Gaussian positions and expanding scene coverage improves multi-view geometric consistency.

Visual Comparisons on Scene Reconstruction. As shown in Figs. 4 and 5, existing methods often suffer from floaters, structural distortions, and blurred textures under the three-view setting. In contrast, ProGCo reconstructs fine textures and accurate structures, particularly in challenging background regions. Its depth map also preserves remarkably correct depth ordering and complete object structures, such as finely and densely structured railings. Under both the 12-view and 24-view settings, ProGCo achieves high-fidelity renderings while retaining intricate background structures, as illustrated in Fig. 6. These results demonstrate the superior reconstruction efficacy of ProGCo across different input-view settings.

C. Ablation Studies

We conduct comprehensive ablation experiments to assess the individual contributions of our core algorithmic components. Additional ablation results are provided in the video demonstration, as described in Section II of the supplementary material. Extended analyses, including evaluations of alternative monocular depth estimators, comparisons with local depth loss variants, and ablations on pseudo-image enhancement, are detailed in Section III of the supplementary material.

Effectiveness of the ProGCo Framework. Table IV demonstrates the efficacy of our progressive geometry-coverage co-refinement framework. Naively incorporating raw

pseudo-views (Exp. (a)) severely degrades the baseline performance, confirming that directly rendering pseudo-images from an initially unreliable point cloud introduces compounding errors into 3DGS optimization. Conversely, removing floaters and applying structural dissimilarity filtering to the rendered pseudo-images (Exp. (b)) effectively improve visual quality and expand spatial coverage. Applying our instance-aware geometric coherence refinement alone (Exp. (c)) yields notable reconstruction gains, but the resulting performance remains suboptimal because of limited spatial coverage. Crucially, as shown in Fig. 8, naively combining these components (Exp. (d)) causes localized structural distortions and the loss of high-frequency details. In contrast, integrating these mechanisms within our progressive geometry-coverage co-refinement framework achieves superior geometric fidelity and rendering accuracy, as illustrated in Figs. 7 and 8. These results demonstrate that prioritizing geometry refinement before expanding scene coverage, followed by their progressive co-refinement, is critical for synthesizing high-fidelity novel views.

Ablation on Geometric Coherence Refinement. As shown in Table V and Fig. 9, instance-aware depth regularization (a) effectively preserves complete object structures and sharp object boundaries. However, it fails to constrain small background structures due to inaccurate segmentation or worse rendered depth in these regions. Concurrently, scene-level relative depth alignment (b) maintains inter-object spatial relationships while recovering these small background objects. Incorporating both constraints (c) yields a substantial performance gain, confirming their complementary roles in establishing reliable

TABLE IV

Ablation of the optimization schedule ON LLFF WITH THREE INPUT VIEWS. “GEOMETRY REFINEMENT” DENOTES THE INSTANCE-AWARE GEOMETRIC COHERENCE REFINEMENT SCHEME.

Exp. (Optimization Paradigm)	Geometry Refinement	Pseudo-view Interpolation	Pseudo-image Enhancement	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Baseline	–	–	–	18.54	0.588	0.272
(a) Standalone Raw Pseudo-View Expansion	–	Random Interpolation	–	17.67	0.496	0.389
(b) Standalone Enhanced Pseudo-View Expansion	–	Direct pseudo-view supervision	✓	19.23	0.623	0.282
(c) Standalone Geometry Refinement	✓	–	–	19.98	0.652	0.194
(d) Naive Combination	✓	Fixed pseudo-view supervision	✓	20.14	0.646	0.297
Ours	✓	Geometry-grounded progressive expansion	✓	22.06	0.740	0.177

TABLE V

Ablation of instance-aware geometric coherence refinement ON LLFF WITH THREE INPUT VIEWS.

Exp.	Instance-aware Con.	Scene-level Align.	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Baseline			18.54	0.588	0.272
(a)	✓		19.24	0.611	0.221
(b)		✓	19.56	0.619	0.213
(c)	✓	✓	19.98	0.652	0.194

TABLE VI

Effect of the number of pseudo-views per interpolation step ON LLFF WITH THREE INPUT VIEWS.

# Pseudo-Views	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
1	21.36	0.719	0.183
2	21.70	0.720	0.194
3	22.06	0.740	0.177

scene geometry.

Ablation on the Number of Pseudo-Views. Table VI evaluates the impact of varying pseudo-view counts (N_{pse}) under the 3-view configuration. Increasing N_{pse} from 1 to 3 consistently improves reconstruction performance, demonstrating that interpolated pseudo-images provide effective supervision for recovering under-observed regions. However, increasing N_{pse} beyond 3 is unnecessary for forward-facing scenes, where the viewpoint range is inherently narrow and adjacent views are densely distributed. Accordingly, we adopt $N_{\text{pse}} = 3$ as the default setting, achieving a favorable balance between additional pseudo-view supervision and the information provided by real observations.

V. CONCLUSION

This paper presents ProGCo, a progressive geometry co-refinement framework for sparse-view 3D Gaussian Splatting. We effectively address the coupled dilemma between inaccurate Gaussian positions and limited spatial coverage. ProGCo first establishes a coherent geometric scaffold across views by preserving intra-object structural integrity and inter-object spatial relationships. Based on the refined geometry, ProGCo reliably interpolates pseudo-views to expand scene coverage. By introducing diffusion-based artifact suppression and structural dissimilarity filtering, the retained pseudo-images provide faithful supervision that facilitates subsequent geometry optimization. This co-refinement mechanism enables high-fidelity NVS. Extensive experiments demonstrate that our

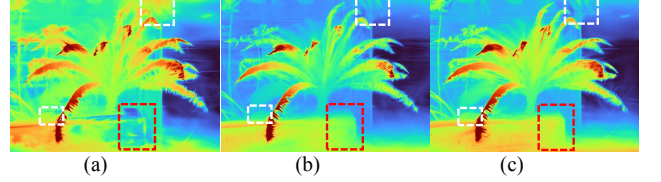


Fig. 9. **Ablation of the two depth constraints.** The instance-level loss preserves complete object structures, while the scene-level loss improves relative depth ordering. Combining both produces the most coherent result.

method consistently achieves superior rendering quality and geometric coherence across diverse sparse-view settings.

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